CH 3 — Function Limits and Continuity

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Definitions

If $f : \mathbb{R} \to \mathbb{R}$ is a function and $a \in \mathbb{R}, \lim_{x \to a} f(x) = L$ if
for all $\varepsilon > 0$ there exists $\delta > 0$ s.t. if $0 < |x - a| < \delta$, then $|f(x) - L| < \varepsilon$

Examples:

1)Prove using the $\varepsilon - \delta$ definition that $\lim_{x \to 0} f(x)$ DNE where

$$f(x) = \begin{cases} -2 \text{ if } x < 0 \\ 3 \text{ if } x > 0 \end{cases}$$

Domain: $\mathbb{R} \setminus \{0\}$

Take $\varepsilon = 1$. Consider some $\delta > 0$. Whitin $(0 - \delta, 0 + \delta)$
We have both $(-\delta, 0)$ where $f(x) = -2$ and $(0, \delta)$ where $f(x) = 3$. If this δ exists for $\varepsilon = 1$ then the limit L would need to be distance 1 or both –2 and 3, where is impossible.

$$\therefore \lim_{x \to 0} f(x) = \text{DNE}$$

2) $\lim_{x \to 7} 8x - 3 = 53$

Let $\varepsilon > 0$ be arbitrary.
We want find δ s.t. if $0 < |x - 7| < \delta$ then $|8x - 3 - 53| < \varepsilon \to \delta = \frac{\varepsilon}{8}$
Pick $\delta = \frac{\varepsilon}{8}$.
Then if $0 < |x - 7| < \frac{\varepsilon}{8}, |(8x - 3) - 53| = |8x - 56| = 8|x - 7| < 8 \cdot \frac{\varepsilon}{8} = \varepsilon$

3) $\lim_{x \to 1} x^2 + 3x + 4 = 8$

We want for any $\varepsilon > 0$ and $\delta > 0 : |x - 1| < \delta$, then $|f(x) - L| < \varepsilon$

$\leftrightarrow |x^2 + 3x - 4| < \varepsilon \leftrightarrow |(x + 4)(x - 1)| < \varepsilon \leftrightarrow |x + 4| - |x - 1| < \varepsilon$

I can always make δ smaller if I need to.

take $\delta < 1$, then $|x - 1| < 1 \implies 0 < x < 2$
$|x + 4| < 6 \to |x + 4 \| x - 1| < 6\delta$, but $6\delta < \varepsilon \leftrightarrow \delta < \frac{\varepsilon}{4}$. Say $\frac{\varepsilon}{7} < \frac{\varepsilon}{6}$ for all epsilon.
Take $\delta < \min\left(1, \frac{\varepsilon}{6}\right)$

Proof

Let $\varepsilon > 0$ be given. Take $\delta = \min\left(\frac{1}{2}, \frac{\varepsilon}{7}\right)$. Then, if $|x - 1| < \delta, |x^2 + 3x + 4 - 8| = |x^2 + 3x - 4| = |(x + 4)(x - 1)| = |(x + 4)(x - 1)| < 6 \cdot \frac{\varepsilon}{7} < \varepsilon$

$\square$

📐 **Theorem — Sequential Characterization of Limits Theorem**
Let $a \in \mathbb{R}$. let the function $f(x)$ be defined on an open interval containing a, expect possibly at $x = a$ itself. Then the following are equivailent:
1. $\lim_{x\to a} f(x) = L$
2. For all sequences $\{x_n\}$ satisfying $\lim_{n\to\infty} x_n = a$ and $x_n \neq a, \forall n \in \mathbb{N}$, we have that $\lim_{n\to\infty} f(x_n) = L$

💡 **Tip** — Usage of Sequential Characterization of Limits
1. Find a sequence $\{x_n\}$ with $x_n \to a$
2. Find two sequences $\{x_n\}, \{y_n\}$ with $x_n, y_n \to a$ and $x_n, y_n \neq a, \forall n \in \mathbb{N}$ but which $\{f(x_n)\}, \{f(y_n)\}$ converge to different values

Proof

$\implies$: $\lim_{x\to a} f(x) = L$ means $\forall \varepsilon > 0, \exists \delta > 0 : |x - a| < \delta$, then $|f(x) - L| < \varepsilon$.
Let $\{x_n\}$ be s.t. $x_n \to a$(meaning that $\forall \varepsilon > 0, \exists N \in \mathbb{R} : \forall n > \mathbb{N}, |x_n - a| < \varepsilon_2$) and $x_n \neq a$ for any n.

In particular, let ε for $x_n \to a$ be δ. Then $\forall n > N$, $|x_n - a| < \delta$, and so $|f(x)_n| < \varepsilon_1$. Then $\forall n > N, |x_n - a| < \delta$ and so $|f(x_n) - L| < \varepsilon_1$. So by definition, $\lim_{n\to\infty} f(x_n) = L$

Side Question: We saw the limit of a sequence is unique. Is the same true for limits of functions?

ANS: NO, it is like saying $\lim_{x\to a} f(x) = L$ and $= M$ and $L \neq M$ Suppose true. By Sequential Characterization of Limits, $\forall \{x_n\} \to a$ but $x_n \neq a \forall n$, $f(x_n) \to L$ and $f(x_n) \to M$ but $L \neq M$ Since the limits of sequences are unique, thus there is a contradiction.

Examples:

Prove that $\lim_{x\to 0} \cos(\frac{1}{x})$ does not exist
We take sequences of peak points of $\cos(\frac{1}{x})$, that is $-1, 1$. Then will converge to $-1, 1$ repeatedly, so by Sequential Characterization, $\lim_{x\to 0} \cos(\frac{1}{x})$ will not exist.

$\cos(\frac{1}{x}) = 1$ if $x = \frac{1}{2k\pi}, k \in \mathbb{Z}$, and $\cos(\frac{1}{x}) = -1$ if $x = \frac{1}{(2k+1)\pi}, k \in \mathbb{Z}$.

Let $x_n = \frac{1}{2}n\pi$ and $y_n = \frac{1}{2n+1}\pi$. Then $x_n, y_n \to 0, x_n, y_n \neq 0 \forall n$. It converges to both -1 and 1. By Sequential Characterization, the limit DNE.

Limit Laws

🐳 **Info** — Let f, g be functions with $\lim_{x\to a} f(x) = L$, $\lim_{x\to a} g(x) = M$ for some $L, M \in \mathbb{R}$ then:
1. For any $c \in \mathbb{R}$, if $f(x) = c$ for all n then $L = c$
2. For any $c \in \mathbb{R}$, if $\lim_{x\to a} cf(x) = cL$
3. $\lim_{x\to a}(f(x) + g(x)) = L + M$
4. $\lim_{x\to} f(x) \cdot g(x) = LM$
5. $\lim_{n\to\infty} \frac{f(x)}{g(x)} = \frac{L}{M}$ if $M \neq 0$
6. If $\alpha > 0$ and $L > 0$, then $\lim_{x>a} f(x)^\alpha = L^\alpha$

Proof

We assume functions f, g are defined on a punctured neighborhood of a and $\lim_{x\to a} f(x) = L$, $\lim_{x\to a} g(x) = M$. In the quotient law we also assume $M \neq 0$.

1. Product law

Claim. $\lim_{x\to a}(f(x)g(x)) = LM$.

Proof. Let $\varepsilon > 0$. Then $|f(x)g(x) - LM| = |f(x)(g(x) - M) + M(f(x) - L)| \leq |f(x)| \cdot |g(x) - M| + |M| \cdot |f(x) - L|$.

Since $f(x) \to L$, choose $\delta_0 > 0$ with $|x - a| < \delta_0 \Rightarrow |f(x) - L| < 1$, hence $|f(x)| \leq |L| + 1$ there.

Choose $\delta_1, \delta_2 > 0$ so that $|x - a| < \delta_1 \Rightarrow |g(x) - M| < \frac{\varepsilon}{2(|L| + 1)}$ and $|x - a| < \delta_2 \Rightarrow |f(x) - L| < \frac{\varepsilon}{2(|M| + 1)}$.

Let $\delta = \min(\delta_0, \delta_1, \delta_2)$. For $0 < |x - a| < \delta$, $|f(x)g(x) - LM| \leq (|L| + 1) \cdot \frac{\varepsilon}{2(|L| + 1)} + |M| \cdot \frac{\varepsilon}{2(|M| + 1)} \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$.

Thus $\lim_{x\to a}(fg) = LM$.

2. Quotient law (with $M \neq 0$)

Claim. $\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{L}{M}$.

Proof. Let $\varepsilon > 0$. Because $g(x) \to M \neq 0$, there exists $\delta_0 > 0$ such that $|x - a| < \delta_0 \Rightarrow |g(x) - M| < |M\frac{|}{2}$, hence $|g(x)| \geq |M\frac{|}{2}$.

Now $|\ \frac{f(x)}{g(x)} - \frac{L}{M}\ | = |\ Mf(x) - Lg(x)\frac{|}{|M| \cdot |g(x)|} \leq \frac{|M| \cdot |f(x) - L| + |L| \cdot |g(x) - M|}{|M| \cdot |g(x)|} \leq \left(\frac{2}{|}M|\right) \cdot |f(x) - L| + \left(2\ |L\frac{|}{|}M|^2\right) \cdot |g(x) - M|$.

Choose $\delta_1, \delta_2 > 0$ with $|x - a| < \delta_1 \Rightarrow |f(x) - L| < \left(|M\frac{|}{4}\right) \cdot \varepsilon$ and $|x - a| < \delta_2 \Rightarrow |g(x) - M| < \left(|M\frac{|^2}{4(|L| + 1)}\right) \cdot \varepsilon$.

Let $\delta = \min(\delta_0, \delta_1, \delta_2)$. Then for $0 < |x - a| < \delta$, $|\ \frac{f(x)}{g(x)} - \frac{L}{M}\ | \leq \left(\frac{2}{|}M|\right) \cdot \left(|M\frac{|}{4}\right) \cdot \varepsilon + \left(2\ |L\frac{|}{|}M|^2\right) \cdot \left(|M\frac{|^2}{4(|L| + 1)}\right) \cdot \varepsilon \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$.

Therefore $\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{L}{M}$.

🐳 **Info — Limit of Polynomial Functions** Let $p(x) = a_0 + a_1x + a_2x^2 + \ldots + a_nx^n$ be a polyomial.

Then $\lim_{x\to a} p(x) = p(a)$

Proof

$$\lim_{x\to a} x = a$$

$$\lim_{x\to a} x^i = a^i$$

$$\lim_{x\to a} a_ix^i = a_ia^i$$

$$\lim_{x\to a} \sum_{i=0}^{n} a_ix^i = \sum_{i=0}^{n} a_ia^i$$

🐳 **Info — Limit of Rational Functions**

Let $f(x) = \frac{p(x)}{q(x)}$ when p, q be polynomial functions and $a \in \mathbb{R}$

1. If $q(a) \neq 0$ then $\lim_{x\to a} \frac{p(x)}{q(x)} = \frac{p(a)}{q(a)}$
2. If $\lim_{x\to a} q(a) = 0$ but then $\lim_{x\to a} p(x) \neq 0$ then $\lim_{x\to a} \frac{p(x)}{q(x)}$ is DNE.
   If $x \to a, x < 0$, then the limit diverges to $-\infty$.
   If $x \to a, x > 0$, then the limit diverges to ∞.
3. Otherwise, $p(a) = 0 = q(a)$, so both $p(x)$ and $q(x)$ have $(x - a)$ as a factor. Divide it out and then repeat the process.

Examples:

1. $\lim_{x\to -3} \frac{x^3+10x^2+13x-24}{x^2-4x-21}$

$\Rightarrow \overset{\left[\frac{0}{0}\right]}{=} \lim_{x\to -3} \frac{(x+3)(x-1)(x+8)}{(x+3)(x-7)} = \lim_{x\to -3} \frac{(x-1)(x+8)}{(x-7)} = \frac{(-3-1)(-3+8)}{(-3-7)} = \frac{-20}{-10} = 2$

📐 **Theorem — Squeeze Theorem(Functions)**:

If $g(x) \leq f(x) \leq h(x)$ be functions defined in an open interval I around a except possibly at a.

If $\forall a \in I \setminus \{a\}$ we have $g(x) \leq f(x) \leq h(x)$ and $\lim_{x\to a} g(x) = \lim_{x\to a} h(x) = L$, then $\lim_{x\to a} f(x) = L$

💡 **Tip** — When to apply Squeeze Theorem

1. Trigonometeric functions with clear bounds and polynomial terms before
2. Exponential Functions with constants terms or by defining a certain inverval

2. Evaluate $\lim_{x\to 0} x^2 \cos\left(\frac{1}{x}\right)$

$$-1 \leq \cos\left(\frac{1}{x}\right) \leq 1$$

$$-x^2 \leq x^2 \cos\left(\frac{1}{x}\right) \leq x^2$$

Notice that x^2 are polynomial function that is defined in $x \in \mathbb{R}$.

$$\lim_{x\to 0} -x^2 = \lim_{x\to 0} x^2 = 0$$

By Squeeze Theorem, $\lim_{x\to 0} x^2 \cos\left(\frac{1}{x}\right) = 0$

One Sided Limits and the Fundamental Trig Limit

1. We say that L is the **right side limit** of f at a, and write $\lim_{x\to a^+} f(x) = L$ if $\forall \varepsilon > 0, \exists \delta > 0$ s.t. if $|x - a| < \delta$ and $x > a$ then $|f(x) - L| < \varepsilon$
2. We say that L is the **left side limit** of f at a, and write $\lim_{x\to a^-} f(x) = L$ if $\forall \varepsilon > 0, \exists \delta > 0$ s.t. if $|x - a| < \delta$ and $x < a$ then $|f(x) - L| < \varepsilon$

📐 **Theorem — Theorem**
$\lim_{x\to a} f(x) = L \iff \lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L$

Example:
Show that $\lim_{x\to 0} \sin(x) = 0, \lim_{x\to 0} \cos(x) = 1,$ and $\lim_{x\to 0} \tan(x) = 0$

1. $\lim_{x\to 0} \sin(x)$:

Construct a unit circle with a triangle that lies in the first quadrant and an arbitrary point, say $P(x, y)$. Then $P(x, y) = P(\cos(x), \sin(y))$. The area of the triangle can be represented as $\frac{1}{2}\sin(x)$.

Contruct another unit circle and draw $P(x, y)$ at the same location as the previous triangle, however, contruct an sector. The area of this new sector is $\frac{1}{2}x$.

Notice that the area bounded by the sector is bigger than the triangle.

We then have $0 \le \frac{1}{2}\sin(x) \le \frac{1}{2}x \implies 0 \le \sin(x) \le x$. Since $\lim_{x\to 0^+} 0 = \lim_{x\to 0^+} x = 0$, by Squeeze Theorem, $\lim_{x\to 0^+} \sin(x) = 0$

$\lim_{x\to 0^-} \sin(x) = 0$ can be achieved similarly to the prove of right side limit and will be omitted.

Thus $\lim_{x\to 0} \sin(x) = 0$

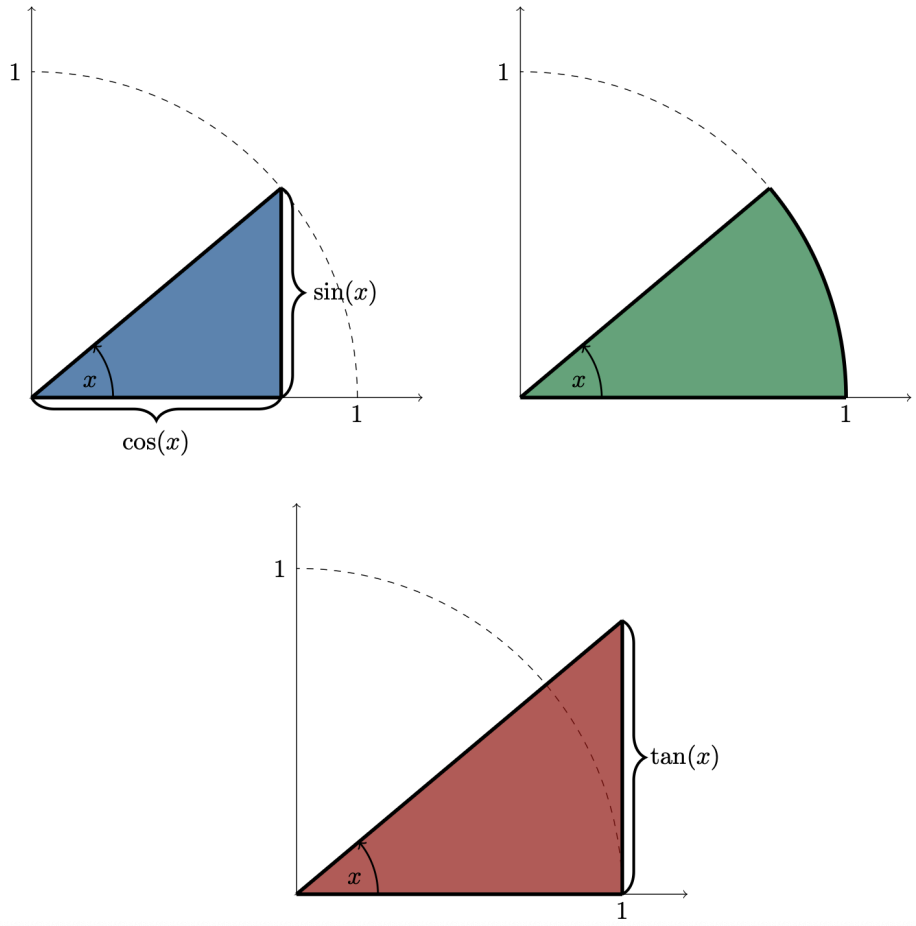
2. $\lim_{x\to 0} \cos(x) = 1$:

$\lim_{x\to 0} \cos(x) = \lim_{x\to 0} \sqrt{1 - \sin^2(x)} = 1$

3. $\lim_{x\to 0} \tan(x) = \lim_{x\to 0} \frac{\sin(x)}{\cos(x)} = 1$

⚠️ **Warning — The Fundamental Trig Limit**:

$$\lim_{x\to 0} \frac{\sin(x)}{x} = 1$$



We have that $\frac{1}{2}\cos(x)\sin(x) \leq \frac{1}{2}x \leq \frac{1}{2}\tan(x) \implies \cos(x) \leq \frac{x}{\sin(x)} \leq \frac{1}{\cos(x)}$.

By Squeeze Theorem, $\lim_{x\to 0^+} \frac{\sin(x)}{x} = 1$.

Since $\sin(x)$ is a even function, then $\frac{\sin(-x)}{-x} = \frac{-\sin(x)}{-x} = \frac{\sin(x)}{x}$ so $\lim_{x\to 0^-} \frac{\sin(x)}{x} = 1$

$$\therefore \lim_{x\to 0} \frac{\sin(x)}{x} = 1$$

Examples:

1. $\lim_{x\to 0} \frac{\tan(x)}{x} = \lim_{x\to 0} \frac{\sin(x)}{x} \cdot \frac{1}{\cos(s)} = \lim_{x\to 0} \frac{\sin(x)}{x} \cdot \lim_{x\to 0} \frac{1}{\cos(x)} = 1 \cdot 1 = 1$
2. $\lim_{x\to 0} \frac{\sin(72x)}{\tan(9x)} = \lim_{x\to 0} \frac{\sin(72x)}{72x} \cdot \frac{9x}{\tan(9x)} \cdot 8 = \lim_{x\to 0} \frac{\sin(72x)}{72x} \cdot \lim_{x\to 0} \frac{9x}{\tan(9x)} \cdot 8 = 1 \cdot 1 \cdot 8$

   $\lim_{x\to 0} \frac{\sin(x^2-1)}{\sin(x-1)} = \lim_{x\to 0} \frac{\sin(x^2-1)}{\sin(x-1)} \cdot \frac{x^2-1}{x^2-1} \cdot \frac{x-1}{x-1} = \lim_{x\to 0} \frac{\sin(x^2-1)}{x^2-1} \cdot \lim_{x\to 0} \frac{x-1}{\sin(x-1)} \cdot$
3. $\lim_{x\to 0}(x+1) = 1 \cdot 1 \cdot 1 = 1$

Horizontal Asymptotes and the Fundamental Log Limit

Info — Limit at $\pm\infty$

Let $L \in \mathbb{R}$.

We say that $\lim_{x\to\infty} f(x) = L$ if $\forall \varepsilon > 0, \exists N \in \mathbb{R}$ s.t. if $x > N$, then $|f(x) - L| < \varepsilon$.

Similarly, $\lim_{x\to-\infty f(x)=L}$ if $\forall \varepsilon > 0, \exists N \in \mathbb{R}$ s.t. if $x < N$, then $|f(x) - L| < \varepsilon$.

Info — Horizontal Asymptotes

If $\lim_{x\to\infty} f(x) = L$ or $\lim_{x\to-\infty} f(x) = L$ for some $L \in \mathbb{R}$ then we way $y = L$ is a **Horizontal Asymptote** of f

Note: you can cross horizontal asymptotes multiple times

Info — Divergence of Limits

1. We say that $\lim_{x\to\infty} f(x) = \infty$ if, $\forall M > 0, \exists N \in \mathbb{R}$ s.t. if $x > N$ we have $f(x) > M$.
2. We say that $\lim_{x\to-\infty} f(x) = \infty$ if, $\forall M > 0, \exists N \in \mathbb{R}$ s.t. if $x < N$ we have $f(x) > M$.
3. We say that $\lim_{x\to\infty} f(x) = -\infty$ if, $\forall M < 0, \exists N \in \mathbb{R}$ s.t. if $x > N$ we have $f(x) < M$.
4. We say that $\lim_{x\to-\infty} f(x) = -\infty$ if, $\forall M < 0, \exists N \in \mathbb{R}$ s.t. if $x < N$ we have $f(x) < M$.

Theorem — Squeeze Theorem at $\pm\infty$

If $g(x) \le f(x) \le h(x) \forall x \ge N$ for some $N \in \mathbb{R}$, and if
$\lim_{x\to\infty} g(x) = L = \lim_{x\to\infty} h(x)$, then $\lim_{x\to\infty} f(x) = L$

Warning — The Fundamental Log Limit

$$\lim_{x\to\infty} \frac{\ln(x)}{x} = 0$$

Proof

$0 \le \frac{\ln(x)}{x}$ true whenever $x \ge 1$. Since $x \to \infty$, assume $x \ge 1$.

$\frac{\ln(x)}{x} = \frac{\ln(\sqrt{x}\cdot\sqrt{x})}{\sqrt{x}\cdot\sqrt{x}} = \frac{2\ln(\sqrt{x})}{\sqrt{x}\cdot\sqrt{x}} = \frac{2}{\sqrt{x}} \cdot \frac{\ln(\sqrt{x})}{\sqrt{x}} \le 1 \le \frac{2}{\sqrt{x}}$ (since $\ln(z) \le z, \forall z$ arbitrarily large)

$0 \le \frac{\ln(x)}{x} \le \frac{2}{\sqrt{x}}$. By Squeeze Theorem $\lim_{x\to\infty} \frac{\ln(x)}{x} = 0$

Examples:

1. Show that $\lim_{x\to\infty} \frac{\ln(x)}{x^p} = 0, \forall p > 0$

$\lim_{x\to\infty} \frac{\ln(x)}{x^p} = \lim_{x\to\infty} \frac{\ln(x^p)}{x^p} \cdot \frac{1}{p}$ Let $u = x^p, x \to \infty, u \to \infty$

$\lim_{u\to\infty} \frac{\ln(u)}{u} \cdot \frac{1}{p} = 0 \cdot \frac{1}{p} = 0$

2. Show that $\lim_{x\to\infty} \frac{\ln(x^p)}{x} = 0, \forall p > 0$

$\lim_{x\to\infty} \frac{\ln(x^p)}{x} = \lim_{x\to\infty} p \cdot \lim_{x\to\infty} \frac{\ln(x)}{x} = p \cdot 0 = 0$

3. Show that $\lim_{x\to\infty} \frac{x^p}{e^x} = 0, \forall p > 0$

$\lim_{x\to\infty} \frac{x^p}{e^x}$ Let $x = \ln u \Leftrightarrow u = e^x, x \to \infty, u \to \infty$
$\lim_{u\to\infty} \frac{\ln(u)^p}{u} = \lim_{u\to\infty} \left(\frac{\ln(u)}{u^{\frac{1}{p}}}\right)^p = 0^p = 0$

4. Show that $\lim_{x\to 0^+} \frac{x^p}{\ln(x)} = 0, \forall p > 0$

$\lim_{x\to 0^+} \frac{x^p}{\ln(x)}$ $\left(\text{Let } u = \frac{1}{x}\right)$
$\lim_{u\to\infty} \frac{1}{u^p} \cdot \ln\left(\frac{1}{u}\right) = 0$

> 🐳 **Info — Vertical Asymptotes**
> 1. $\lim_{x\to a^+} f(x) = \infty$ if for every $m > 0, \exists \delta > 0$ s.t. if $a < x < a + \delta$ then $f(x) > m$
> 2. $\lim_{x\to a^-} f(x) = \infty$ if for every $m > 0, \exists \delta > 0$ s.t. if $a - \delta < x < a$ then $f(x) > m$
> 3. $\lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x) = \infty \implies \lim_{x\to a} f(x) = \infty$
> 4. If $\lim_{x\to a^\pm} f(x) = \pm\infty$ then there is a vertical asymptote at $x = a$

> 🐳 **Info — Continuity**
> We say that a function f **is continuous** at a if f is defined at a and
> 1. $\lim_{x\to a} f(x) = f(a)$
> 2. For every $\varepsilon > 0, \exists \delta > 0$ s.t. if $|x - a| < \delta$, then $|f(x) - f(a)| < \varepsilon$
> 3. for every sequence $\{x_n\}$ with $x_n \to a$, we have $f(x_n) \to f(a)$
> 4. If f is not continuous at a, then it is called **discountinous at** a

Example:
1. $f(x) = |x|$ is continuous at a

$f(x) = \begin{cases} x \text{ if } x>0 \\ -x \text{ if } x\leq 0 \end{cases}$

$\lim_{x\to 0^-} |x| = \lim_{x\to 0^-} -x = 0$
$\lim_{x\to 0^+} |x| = \lim_{x\to 0^+} x = 0$
Thus $\lim_{x\to 0} |x| = 0$

> 💡 **Tip — Polynomials are continuous at all points**
> $$\lim_{x\to a} p(x) = p(a)$$

> 🐳 **Info** — Alternative way of Proving Continuity
> $$\lim_{x\to a} f(x) = f(a) \equiv \lim_{h\to 0} f(a+h) = f(a)$$

Example:
1. Prove $\sin(x)$ is continuous at $x \in \mathbb{R}$

$\sin(a + h) = \sin(a)\cos(h) + \sin(h)\cos(a)$
$\lim_{x\to a} \sin(x) = \lim_{h\to 0} \sin(a + h)$
$= \lim_{h\to 0} \sin(a) \cdot \lim_{h\to 0} \cos(h) + \lim_{h\to 0} \sin(h) \cdot \lim_{h\to 0} \cos(a)$
$= \sin(a) \cdot 1 + 0 \cdot \cos(a) = \sin(a)$

2. Given that e^x is continuous at $x = 0$, prove that it is continuous on all of $\mathbb{R}$

$\lim_{x\to a} e^x = \lim_{h\to 0} e^{a+h} = e^a \cdot 1 = \lim_{h\to 0}$

> 💡 **Tip — Continuity of Inverse**
> Let $f(x)$ be invertible at $x = a$ with $f(a) = b$. Then $f^{-1}(y)$ is continuous at $y = b$

> 🐳 **Info — Arithmetic Rules for Continuous Functions**
> Let f, g be continuous at x = a.
> 1. For any constant $c \in \mathbb{R}$, cf(x) is continuous at $x = a$
> 2. $f(x) \pm g(x)$ remain continuous
> 3. $f(x)g(x)$ remain continuous
> 4. If $g(x) \neq 0$, $\frac{f(x)}{g(x)}$ remain continuous

> 🐳 **Info — Composition of Continuous Function**
> If f is continuous at $x = a$, and g is continuous at $x = f(a)$, then
> $h = g \circ f$ is continuous at $x = a$

Proof:
Given f is continuous at a, then for any $\{x_n\}, x_n \to a, f(x_n) \to f(a)$ But g is continuous at $f(a)$, so for any $\{y_n\}$ with $y_n \to f(a), g(y_n) \to g(f(a))$. Let $y_n = f(x_n)$. Then $g(f(x_n)) \to g(f(a))$ since x_n is aribitrary, $g \circ f$ is continuous at $x = a$

> 💡 **Tip — Continuity on Invervals**
> 1. We say that a function f is continuous on (a, b) if it is continuous at each $x \in (a, b)$
> 2. We say that a function f is continuous on $[a, b]$ if it is continuous on (a, b) and we have $\lim_{x\to a^-} f(x) = f(a)$ and $\lim_{x\to b^+} f(x) = f(b)$

Example:
Find the largest interval on which $f(x) = x^{\frac{1}{4}}$ is continuous.
For $(0, \infty)$ given $a \in (0, \infty)$, f is continuous at a^- beacuse it is the inverse of x^4, which is continuous at $\mathbb{R}$.

For $a = 0, \lim_{x\to 0^+} x^{\frac{1}{4}} = 0$, so f is continuous at 0 as well.

$\therefore f$ is continuous on $[0, \infty)$

Info — Types of Discountinuity

1. A discontinuity is a removable discontinuity if $\lim_{x\to a} f(x)$ exists but does not equal $f(a)$. All discontinuities not of this type are non-removable.
2. A discontinuity is a jump discontinuity if $\lim_{x\to a^-} f(x)$ and $\lim_{x\to a^+} f(x)$ exist but do not agree.
3. A discontinuity is infinite if either $\lim_{x\to a^-} f(x) = \pm\infty$ or $\lim_{x\to a^+} f(x) = \pm\infty$.
4. A discontinuity is oscillatory if $\lim_{x\to a} f(x)$ does not exist, but f is bounded and oscillates infinitely often near $x = a$.

Theorem — Intermediate Value Theorem (IVT)
If f is continuous on a closed interval $[a, b]$ and $\alpha \in \mathbb{R}$ is such that either $f(a) < \alpha < f(b)$ or $f(a) > \alpha > f(b)$, then $\exists c \in (a, b)$ s.t. $f(c) = \alpha$

Examples:
1. Show that $\sin(x)$ and $1 - x^2$ intersects on $(0, 1)$

Since $\sin(x)$ and $1 - x^2$ are continuous on $\mathbb{R}$ therefore continuous on $[0, 1]$
For $x = 1, \sin(1) - 1 + (1)^2 = 1^2 - 1 > 1$
For $x = 0, \sin(0) - 1 + (0)^2 = -1 < 1$
$f(0) < 0 < f(1)$
By IVT, $\exists c \in (0, 1), f(c) = 0$, meaning the given function intersect

2. Show that $x^3 + 3x^2 - x - 3$ has a root in the interval $(-5, -2)$

Since $x^3 + 3x^2 - x - 3$ is continuous on $\mathbb{R}$ therefore continuous on $[-5, -2]$
For $x = -5, (-5)^3 + 3(-5)^2 - (-5) - 3 = -48 < 0$
For $x = -2, (-2)^3 + 3(-2)^2 - (-2) - 3 = 3 > 0$
$-48 < 0 < 3$
By IVT, $\exists c \in (-5, -2), f(c) = 0$, meaning there is a point where the given function has a root

Info — Bisection Method
Let F be a continuous function.
To approximate a value c s.t. $F(c) = 0$ with an error of at most ε

1. We need to find a, b s.t. $F(a)$ and $F(b)$ have opposite signs
2. IVT gives that $c \in (a, b)$
3. Look at the mid-point $a + \frac{b-a}{2}$. Is $F\left(a + \frac{b-a}{2}\right)$ positive or negative?
   $\begin{cases} 1. \text{ If it has the same sigh as } F(a), \text{ then it has the opposite sign with } F(b) \Longrightarrow \text{IVT gives } c \in (a + \frac{b-a}{2}, b) \\ 2. \text{ If it has the opposite sigh as } F(a), \text{ then it has the same sign with } F(b) \Longrightarrow \text{IVT gives } c \in (a, a + \frac{b-a}{2})) \end{cases}$
4. Repeat Step 3 until $|b - a| < \varepsilon$

CHAPTER ENDS